

TECHNICAL REPORT: AIRFLOW SIMULATION

Project : Batiment_R+4

Client : Promoteur_X

Location : Lyon_69

1. NETWORK SUMMARY

Category	Parameter	Value
Air properties	Temperature used (°C)	20.00
	Altitude (m)	170.00
	Density (kg/m³)	1.180
	Dynamic viscosity (Pa·s)	1.813e-05
General metrics	Ventilation type	Supply
	Design volumetric flow rate (m³/h)	2200.00
Fan: fan_01	Fan label	Main_Fan
	Description	Fresh air supply
	Nominal target efficiency (%)	75%
	Fan flow (m³/h)	2200.00
	Critical node	of_03
	Total pressure (Pa)	36.03
	Shaft input power (W)	29.36
Operating cost projections	Total shaft input power (W)	29.36
	Daily operating cycle (h/day)	10.00
	Annual operating cycle (days/year)	250.00
	Annual energy usage (kWh)	73.40
	Electricity Unit Price (€/kWh)	0.25
	Estimated annual opex (€)	18.35

2. TERMINAL BALANCING ANALYSIS

Terminal Node	Total Press. (Pa)	Balance Press. (Pa)	Exit Dyn. Pre. (Pa)	K-Factor
of_01	32.46	3.57	19.57	0.183
of_02	31.72	4.31	18.47	0.233
of_03	36.03	0.00	20.62	0.000
of_04	35.87	0.16	19.57	0.008

3. DUCT PERFORMANCE ANALYSIS

Duct Name	Flow Rate (m³/h)	Velocity (m/s)	Linear Loss (Pa)	Lin. Loss/m (Pa/m)
main_trunk	2200.00	3.84	0.75	0.373
to_junction_A	2200.00	4.86	2.01	0.669
office_1	550.00	4.49	2.72	1.359
office_2	500.00	4.34	2.66	1.330
transit_to_B	1150.00	4.52	2.49	0.830
office_3	600.00	4.63	2.76	1.381
office_4	550.00	4.49	2.72	1.359

4. HYDRAULIC AUDIT DETAILS

Duct Name	Reynolds	Friction Lambda	Sing. Loss (Pa)	Zeta Total
main_trunk	112523	0.0193	0.00	0.000
to_junction_A	126588	0.0192	0.23	0.016
office_1	53741	0.0210	26.76	2.246
office_2	50213	0.0213	26.09	2.347
transit_to_B	88228	0.0207	7.60	0.630
office_3	57084	0.0207	20.20	1.598
office_4	53741	0.0210	20.08	1.686

5. ACOUSTIC NOISE RISK ASSESSMENT

Duct Name	Topology Type	Velocity (m/s)	Noise Lw [dB(A)]	Acoustic Status
main_trunk	Main Trunk	3.84	31.20	OPTIMAL
to_junction_A	Intermediary	4.86	35.30	OPTIMAL
office_1	Terminal Runout	4.49	27.90	OPTIMAL
office_2	Terminal Runout	4.34	26.90	OPTIMAL
transit_to_B	Intermediary	4.52	31.20	OPTIMAL
office_3	Terminal Runout	4.63	28.80	OPTIMAL
office_4	Terminal Runout	4.49	27.90	OPTIMAL

6. ACOUSTIC SIZING RECOMMENDATIONS (CRITICAL)

Duct Name	Status	Vitesse (m/s)	Circ. Diam (mm)	Rect. WxH (mm)
No critical ducts identified				

7. SOLVER METADATA

Performance Metric	Result
Execution time (s)	0.188
Convergence status	stable
Residual error (m³/s)	8.5e-10
Tolerance targeted (m³/s)	1e-09
Iterations performed	622
Max iterations allowed	1000000
Timestamp	08/06/2026 00:29:19

8. NETWORK DIAGRAM

